

Session 3 · Deck 3.A2

Value Function Iteration: The First Complete Crossing

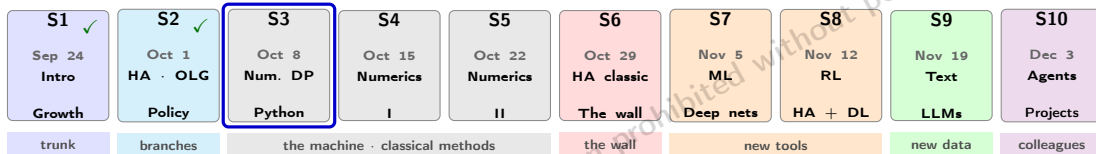
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Build it on paper → solve it on the machine → push it to the frontier with AI

Where We Are

you are here: 3.A2 ▼



Build it on paper → solve it on the machine → push it to the frontier with AI

Block A: 3.A1 what it means to solve a model · 3.A2 value function iteration on a grid · 3.A3 Euler equations and time iteration · 3.A4 the worked example, end to end.

Hands-on, 3.B5:

1 Version 0: grid VFI against the closed form

2 Data → parameters: $g, n, \alpha, \delta, \beta$

3 The calibrated economy, solved and checked

4 Test the model: untar-getted ratio, Euler residual

VFI on a Grid, and the Number It Has to Hit

Input: grid $k_1 < \dots < k_N$ ($N = 200$ on $[0.05, 0.60]$); $\alpha = 0.36$, $\beta = 0.95$; $\varepsilon = 10^{-6}$.

- 1: $U_{ij} \leftarrow \ln(k_i^\alpha - k_j)$ if $k_i^\alpha > k_j$, else -10^{10} ▷ built once
- 2: $v \leftarrow 0$, $n \leftarrow 0$
- 3: **repeat**
- 4: **for each** i **do**
- 5: $v_i^{\text{new}} \leftarrow \max_j \{U_{ij} + \beta v_j\}$; $j_i^* \leftarrow$ the maximizing j
- 6: **end for**
- 7: $d \leftarrow \max_i |v_i^{\text{new}} - v_i|$; $v \leftarrow v^{\text{new}}$; $n \leftarrow n + 1$
- 8: **until** $d < \varepsilon$
- 9: $g(k_i) \leftarrow k_{j_i^*}$ ▷ k' never leaves the grid
- 10: **return** n and $\max_i |g(k_i)/(\alpha\beta k_i^\alpha) - 1|$ ▷ graded

Every decision of 3.A1 is in those ten lines, each made the crudest way available — Decision 3 by not interpolating at all.

Run it

272 sweeps, and a policy that is wrong by up to **0.99%**.

In Python this is twenty lines: Lab L03 runs them, and 3.B2 rebuilds them as a class.

The rest of this deck is two questions: **where did the 1% come from**, and **why 272?**

What Those Lines Actually Do

3.A1 decision	The crudest choice	line
1 Represent v	N numbers v_i	2
2 The grid	uniform, $N = 200$	input
3 Off the grid	never: every k' is a grid point	5, 9
4 The choice	compare all N candidates	5
5 Expectation	none yet: no shock (end of deck)	—
6 Stop	$\max_i v_i^{\text{new}} - v_i < \varepsilon$ ●	7–8

The one trick worth naming

U does not depend on v , so it is built **once** and reused 272 times. Infeasible cells get a large negative number, not $-\infty$, and are not tested inside the loop.

The max runs over *grid points*, so the policy is forced onto the grid. That single decision is the source of most of the 1%.

Decision 2: Which Points, and How Many?

Uniform. $k_i = a + (i - 1) \frac{b-a}{N-1}$. Simple, but it spends points evenly where the error does not fall evenly: interpolation error scales with curvature, and v is far more curved at low k .

Logarithmic. $k_i = a + (b - a) \frac{e^{\theta(i-1)/(N-1)} - 1}{e^\theta - 1}$, with $\theta \in [2, 5]$: points bunched where the function bends.

Both cost the same to evaluate. The second is almost always better, and almost always forgotten.

Two boundary mistakes

Too narrow and the policy hits the top of the grid: $g(k_N) = k_N$ is a censored answer, not a solution.

Too wide and most of your points sit where the model never visits, paying $O(N^2)$ for nothing.

Always print $\max_i g(k_i)/k_N$. If it is 1, widen the grid before you believe anything.

Decision 3: Off the Grid, Straight Lines Only

Between two grid points,

$$\hat{v}(k) = v_i + \frac{v_{i+1} - v_i}{k_{i+1} - k_i} (k - k_i).$$

- Error is $O(h^2)$, with h the spacing.
- It **preserves monotonicity and concavity** — so a concave v stays concave, and the maximization stays well behaved.
- Its derivative is discontinuous at every node. Anything that differentiates $\hat{v}$ will feel those kinks.

Why not something smoother?

Cubic splines are $O(h^4)$ and C^2 — and they **overshoot**, which can destroy concavity and send the maximizer somewhere absurd.

Shape preservation beats formal accuracy inside a fixed-point loop. We stay linear today; the shape-preserving alternatives are built next week.

Decision 6: One Theorem, Used Two Ways

The contraction theorem gives *two* inequalities, and they answer different questions.

How many iterations? From v_0 ,

$$\|v_n - v^*\| \leq \frac{\beta^n}{1 - \beta} \|v_1 - v_0\| \Rightarrow n \gtrsim \frac{\ln \varepsilon}{\ln \beta}.$$

How wrong am I now? From the last step,

$$\|v_n - v^*\| \leq \frac{\beta}{1 - \beta} \|v_n - v_{n-1}\|.$$

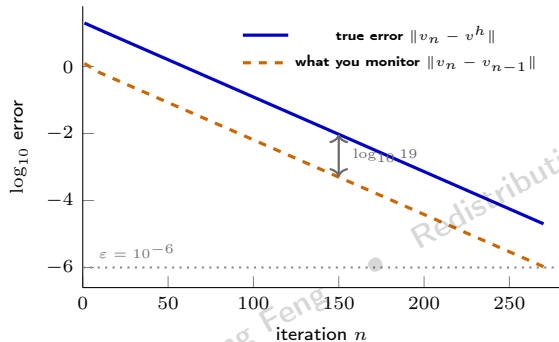
At $\beta = 0.95$ that factor is **19**.

$\beta^n / (1 - \beta)$ from the contraction mapping theorem; the one-step form is 1.B2. Measured counts at $\varepsilon = 10^{-6}$, $N = 200$.

β	$\ln \varepsilon / \ln \beta$	measured	$\frac{\beta}{1 - \beta}$
0.90	131.1	133	9
0.95	269.3	272	19
0.99	1374.6	1379	99

The count is a property of β alone — not of the grid, not of the model. Refining from $N = 50$ to $N = 800$ changed the iteration count by **zero**.

The Stopping Rule Is Not the Error



The two lines are **parallel**, separated by exactly $\log_{10} 19$. The quantity you watch is always a constant factor below the quantity you care about.

At $n = 272$, when the loop stops:

$\ v_n - v_{n-1}\ $	9.6×10^{-7}
guaranteed bound, $19\times$	1.81×10^{-5}
true error	1.81×10^{-5}

The $19\times$ bound is not pessimistic here — it is **attained**. Ask for 10^{-6} and you get 1.8×10^{-5} .

A Better Stopping Rule (1): Trap v^h Between Two Numbers

After sweep n you hold v_n and the change $d_n = v_n - v_{n-1}$ at every grid point. The naive rule keeps only $\max |d_n|$. Keep instead the **smallest** and **largest** change, $d_{\min}$ and $d_{\max}$.

Two facts about T : it is **monotone** ($v \geq w \Rightarrow Tv \geq Tw$) and **discounts** a constant ($T(v + c) = Tv + \beta c$).

1. By definition, $d_{\min} \leq v_n - v_{n-1} \leq d_{\max}$ at every point.

2. Apply T to both sides: $\beta d_{\min} \leq v_{n+1} - v_n \leq \beta d_{\max}$.

3. Repeat: the change j sweeps ahead lies in $[\beta^j d_{\min}, \beta^j d_{\max}]$.

4. v^h is v_n plus *all* the changes still to come. Sum the geometric series, $\beta + \beta^2 + \dots = \frac{\beta}{1-\beta} = 19$:

$$v_n + 19 d_{\min} \leq v^h \leq v_n + 19 d_{\max} \quad \text{at every grid point.}$$

The idea in one sentence

Once every grid point changes by about the same amount, the remaining sweeps only repeat that change, shrinking by β each time — so add them up *now* instead of waiting for them.

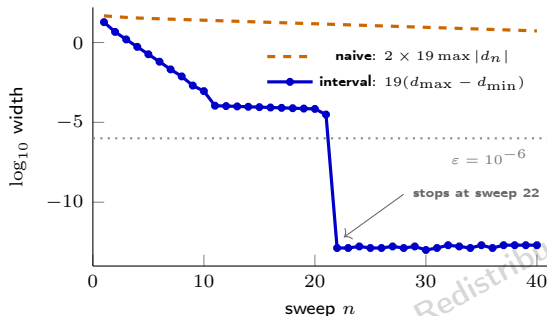
Sweep 5 on our grid

Every value fell by between 0.8425 and 0.8522. So v^h lies in $[v_5 - 16.19, v_5 - 16.01]$: an interval of width **0.18**.

The naive rule only says $|v^h - v_5| \leq 19 \times 0.8522 = 16.2$: an interval of width 32.4.

MacQueen (1966); Porteus (1971). v^h is the fixed point on the grid, as on the previous frame. Ledger: stopping_detail.

A Better Stopping Rule (2): Stop at Sweep 22, Not 272



The naive rule ($\max |d_n| < \epsilon$) fires only at sweep 272.

	stops at	error
naive rule, report v_n	sweep 272	1.8×10^{-5}
interval, report midpoint	sweep 22	9×10^{-14}

Two uses, no extra sweeps

The sweep already computes d_n everywhere; only its minimum is new.

Estimate: report the interval's *midpoint*, not v_n .

Stop: when the width $19(d_{\max} - d_{\min}) < \epsilon$.

Why the gain is so large here

At sweep 22, v_{22} itself is still off by 6.7, yet the midpoint is exact to rounding. Here every capital path leads to the same steady state, so once the policy settles all points change alike: $d_{\min} \approx d_{\max}$. Usually the gain is smaller — but the interval always lies *inside* the naive one.

What It Costs

Each sweep compares every grid point against every candidate choice: $O(N^2)$ per iteration, $O(N^2 N_z)$ with a shock once Ev is precomputed. The iteration count is $O(\ln \varepsilon / \ln \beta)$ and does not depend on N . So the total is

$$\underbrace{N^2}_{\text{grid}} \times \underbrace{\frac{\ln \varepsilon}{\ln \beta}}_{\text{patience}} .$$

N	sweeps	comparisons
200	272	10.9 million
400	272	43.5 million
800	272	174 million

Two ways this becomes fatal

Patience. At a quarterly $\beta = 0.99$ the sweep count is 1,379 instead of 272 — a $5\times$ bill for changing the calendar.

Dimension. A second continuous state makes the grid N^2 and the sweep N^4 . 200 points per dimension is 1.6×10^9 comparisons *per sweep*.

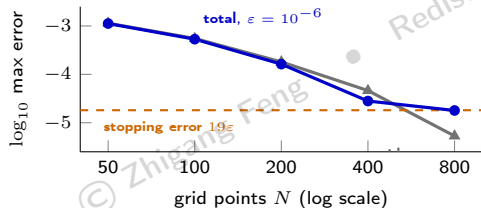
Nothing here is wasted motion — it is the crude choices being paid for. Every one has a cheaper replacement, and that is next week.

Refining the Grid: Two Errors, and the Larger One Wins

What you report, v_n , misses v^* for two reasons:

$$v_n - v^* = \underbrace{(v^h - v^*)}_{\text{grid error}} + \underbrace{(v_n - v^h)}_{\text{stopping error}}.$$

- **Grid error.** k' must sit on the grid, so $v^h \leq v^*$. It shrinks like h^2 (measured slope **1.89**); the policy error like h (**0.93**).
- **Stopping error.** Quitting at sweep 272 leaves $+1.8 \times 10^{-5} = 19\epsilon$ at every point, whatever N .



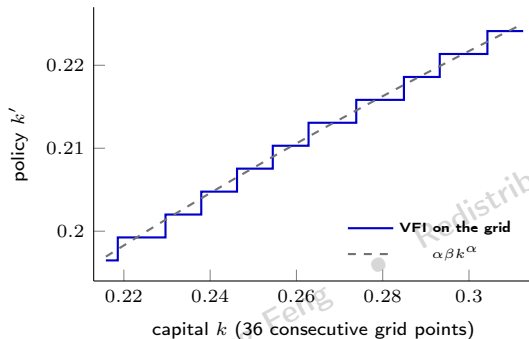
N	grid	stopping	total	work
200	1.8×10^{-4}	1.8×10^{-5}	1.6×10^{-4}	$1 \times$
400	4.6×10^{-5}	1.8×10^{-5}	2.8×10^{-5}	$4 \times$
800	5.3×10^{-6}	1.8×10^{-5}	1.8×10^{-5}	$16 \times$

The two errors have **opposite signs**, so they partly cancel: at $N = 400$, $4.6 - 1.8 = 2.8$.

Where refining stops helping

At $N = 800$ the grid error has fallen *below* the stopping error, so the total is just the stopping error. The last $4 \times$ of work bought almost nothing. To go further, tighten ϵ *and* refine the grid.

Graded Against the Closed Form



Zoom in and the agreement becomes a **staircase**. Across these 36 states the computed policy takes only **11 distinct values** — it can return nothing but a grid point.

$\max v - v^* $	1.6×10^{-4}
max relative policy error	0.99%
grid spacing as a share of k'	$\approx 1.4\%$

The error is a **sawtooth**, not noise, and it does not shrink with more iterations. It is decision 4 — searching the grid — and no amount of patience will fix it.

Decision 5: Where Does the Expectation Come From?

Add productivity $z \in \{z_1, \dots, z_{N_z}\}$ with transition matrix Π . The sweep becomes

$$v_{n+1}(k_i, z_j) = \max_l \left\{ U_{ijl} + \beta \sum_m \Pi_{jm} v_n(k_l, z_m) \right\},$$

and the expectation is a matrix product: $EV = V @ \Pi.T$, computed once per sweep rather than inside the maximization.

With $z \in \{0.95, 1.05\}$ and Π symmetric with persistence 0.9, the closed form still holds state by state — so we can still grade it.

The chain used here is the one introduced in 1.B1.

But where did Π come from?

We wrote it down. Real work starts from an estimated AR(1), $\ln z' = \rho \ln z + \sigma \epsilon'$, and must turn it into a chain that matches the persistence and the variance.

That is a genuine gap, not a detail: the two standard constructions disagree badly when $\rho > 0.95$, which is exactly the range macro uses. We build both next week.

Six Crude Choices, Six Gaps

What we did	What it cost	What we need	
Searched the grid for the max	the 1% sawtooth; $O(N^2)$ per sweep	optimization on an interval	S4
Linear interpolation (3.A3)	$O(h^2)$, kinked derivatives	approximation theory	S4
Wrote Π down by hand	no link to any estimated process	discretizing an AR(1)	S4
Naive stopping rule $N^2 \times 272$ comparisons	19× the error you asked for 174 million at $N = 800$	two-sided bounds acceleration, better operators	today S4
One state variable	nothing yet — and that is the problem	perturbation, projection	S5

Every row is a numerical technique that exists **because** a crude choice hurt. That is the honest reason the next two sessions look like a mathematics course: each tool on the list was invented to pay off one of these lines.

Takeaways

- A loop of ten lines takes you from the Bellman equation to a policy function. The crossing is **not** the hard part — knowing how wrong you are is.
- The iteration count belongs to β alone: 272 sweeps at $\beta = 0.95$, 1,379 at 0.99, whatever the grid. The **accuracy** belongs to the grid.
- The quantity you monitor, $\|v_n - v_{n-1}\|$, sits a factor $\beta/(1 - \beta)$ below the error you care about — $19\times$ here, and **attained**. Keep the smallest change as well and the same sweeps give an **interval**: stop at 22, not 272.
- Error budgets compose: grid error plus stopping error, and the larger one wins. At $\varepsilon = 10^{-6}$, doubling the grid — four times the work — barely moved the value error, because the stopping error was already the larger.
- The 1% policy error is a **sawtooth**, not noise: $\arg \max$ can only return a grid point. Diagnose the shape of an error, not just its size.

Bridge: The Same Model, Down the Other Road

Value function iteration never used the first-order condition. It maximized, brutally, at every point of every sweep — and the price was a policy quantized to the grid.

But we know the optimum satisfies

$$u'(c) = \beta \mathbb{E}[u'(c')R(k', z')].$$

That is an equation the policy must satisfy *exactly*, and solving an equation is a different — often cheaper, often sharper — numerical problem than searching for a maximum.

Next (3.A3)

Iterate on the **policy** instead of the value: the Coleman and Greenwood–Huffman operators, one root-find per grid point.

Then the comparison that matters — two methods, one model, measured against the same closed form — and the list of what still breaks when the model gets harder.